C 20660 **(Pages : 2)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS—UG)

Physics/Applied Physics

PHY 6B 12/APH 6B 12—NUCLEAR PHYSICS AND PARTICLE PHYSICS

(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A (Short Answer Type)

Answer at least **eight** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 24.

1. Discuss any *two* properties of nuclear forces.
2. Draw a typical binding energy per nucleon versus mass number plot indicating its features.
3. Explain the Gamma decay process.
4. Distinguish between prompt and delayed neutrons.
5. What are the basic requirements for a plasma state ?
6. What is the working principle of a proportional counter ?
7. What is the working principle of a Van de Graaff accelerator ?
8. Draw the schematic of a proton synchrotron.
9. What is the working principle of an intersecting beam accelerator ?
10. Give the list of leptons. Mention the charge of each particle.
11. What is the strange behavior of kaons and hyperons ?
12. What do you mean by quantum chromodynamics ?

(8 × 3 = 24 marks)

Turn over

Section B (Paragraph/Problem Type)

Answer at least **five** *questions.*
Each question carries 5 *marks.*
All questions can be attended.
Overall Ceiling 25.

13. Estimate the binding energy of the nucleus $^{12}_{6}C$. Also determine its density.
14. With the help of a neat diagram, explain the working principle of a Wilson cloud chamber.
15. Briefly explain the working principle of a proportional counter using a suitable figure.
16. Using a suitable figure, discuss the principle of a betatron accelerator.
17. Determine the energy released when three alpha particles combine to form ^{12}C.
18. Find the Q value of the following decay.

    $\pi^- \rightarrow \mu^- + \nu_\mu$

19. Name the conservation law that would be violated in the following decay.

    $\nu_e + p \rightarrow e^+ + n.$

(5 × 5 = 25 marks)

Section C (Essay Type)

Answer any **one** *question.*
The question carries 11 *marks.*

20. Explain the conservation laws in radioactive decays using suitable examples.
21. Discuss the stages involved in nuclear fission illustrating examples. Explain the basic features of fission reactions that make it useful as a means to generate electrical energy.

(1 × 11 = 11 marks)